INT421—Digital Signal Processing

Basic Sequences and Operations

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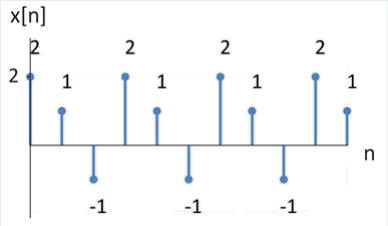


Topics of Last Lecture

- Fundamental signals
  - Unit sample, unit step, exponential, sinusoid
- Basic Signal Operations
  - shifting, folding, etc.
- System representation
  - A functional module between an input signal and an output signal

Quiz

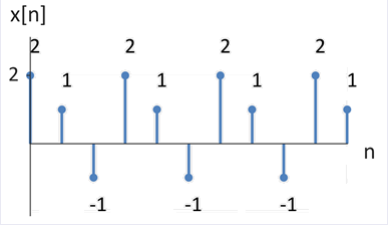
Examples



1. Compute the energy of the signal.
2. Compute the power of the signal

Quiz

Solution



1. *Compute the energy of the signal* $E = \sum_{-\infty}^{\infty} |x[n]|^2 = \infty$
2. *Compute the power of the signal, p for periodic signals, this simplifies to:* $P = \frac{1}{N} \sum_{n=0}^{N-1} |x[n]|^2 = \frac{1}{3}[2^2 + 1^2 + (-1)^2] = 2$ *where N is a period of the signal.*

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2. Basic Signal Operations
3. Time Scaling Operation
4. Complex Sequences
5. Summary

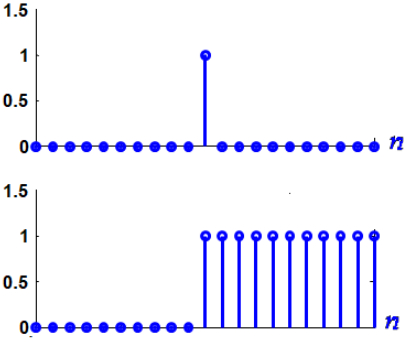
Basic Sequences

- Standard forms of some discrete-time signals that are frequently used in DSP are:
  1. **Impulse**: The simplest signal is the unit impulse signal which is defined as,

$$\delta[n] = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases}$$

  2. **Step**: The general form of step function is,

$$u[n] = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases}$$



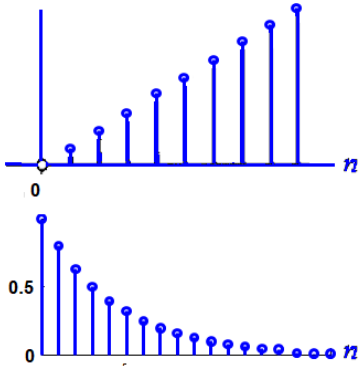
Basic Sequences

- Standard forms of some discrete-time signals that are frequently used in DSP are:
  1. **Ramp**: This signal is given by,

$$r[n] = \begin{cases} n, & n \geq 0 \\ 0, & n < 0 \end{cases}$$

  2. **Exponential**: The decaying exponential is given by,

$$s[n] = a^n \quad \forall n$$



Example

- The unit impulse/sample δ takes the value of 1 whenever its argument (contained in []) **becomes zero**.
- for example:

$$\delta[3n-6] = \begin{cases} 1, & n = 2 \\ 0, & n \neq 2 \end{cases} \quad n \in \mathbb{Z}$$

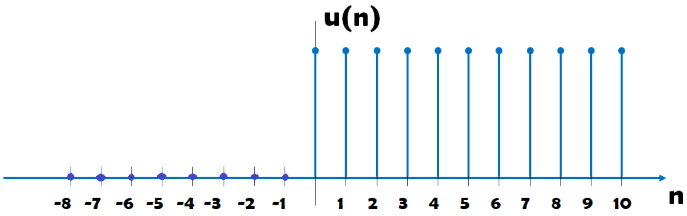
Basic Sequences

Example

- The unit-step sequence $u[n]$

$$u[n] = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases} = \delta[n] + \delta[n-1] + \delta[n-2] + \cdots = \sum_{k=0}^{\infty} \delta[n-k]$$

$$\delta[n] = u[n] - u[n-1]$$



Basic Sequences

Example

- The unit step u takes the value of 1 whenever its argument (contained in []) **becomes zero**.
- for example:

$$u[2n-6] = \begin{cases} 1, & n \geq 3 \\ 0, & n < 3 \end{cases} \quad n \in \mathbb{Z}$$

Basic Sequences

Example

- Unit ramp function $r[n]$ can be expressed as function of unit sample function

$$r[n] = \begin{cases} n, & n \geq 0 \\ 0, & n < 0 \end{cases} = r[n] = nu[n]$$

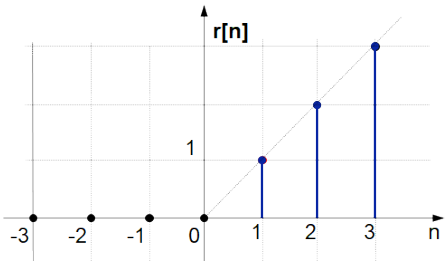


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2 Basic Signal Operations

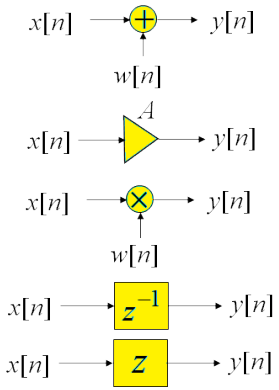
3 Time Scaling Operation

4 Complex Sequences

5 Summary

Operations on Sequences

- Addition operation
  - Adder $y[n] = x[n] + W[n]$
- Multiplication operation
  - Multiplier $y[n] = A.x[n]$
- Product (modulation) operation
  - Modulator $y[n] = x[n].w[n]$
  - Windowing: multiplying an infinite-length sequence by a finite-length window sequence to extract a region.
- Time shifting $y[n] = x[n - N]$, where N is an integer.
  - If $N > 0$, it is delaying operation
    - Unit delay: $y[n] = x[n - 1]$
  - If $N < 0$, it is an advance operation
    - Unit advance: $y[n] = x[n + 1]$
- Reflection or Folding $y[n] = x[-n]$



Operations on Sequences

Example

Suppose the original sequence $x[n] = \{\underset{\uparrow}{1},\ 2,\ 3,\ 4,\ 5\}$

Solution

1. *Time delay*

$$x[n-2] = \{\underset{\uparrow}{0},\ 0,\ 1,\ 2,\ 3,\ 4,\ 5\}$$

2. *Time advance*

$$x[n+2] = \{\ 1,\ 2,\ \underset{\uparrow}{3},\ 4,\ 5\}$$

Operations on Sequences

Solution

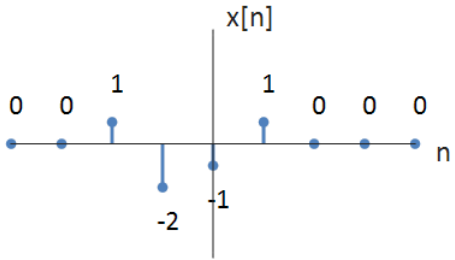
1. *Folding*

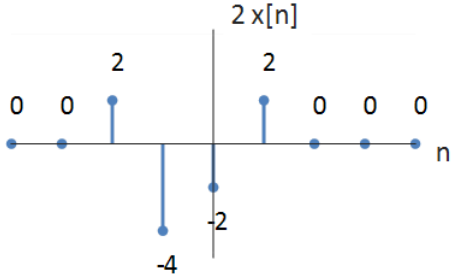
$$x[-n] = \{5,\ 4,\ 3,\ 2,\ \underset{\uparrow}{1}\}$$

2. *Folding and delay* $x[-(n-k)] = x[-n+k]$.
   - *First fold the sequence* $x[n]$*; that means obtain* $x[-n]$
   - *Then delay the folded sequence by k sample.*

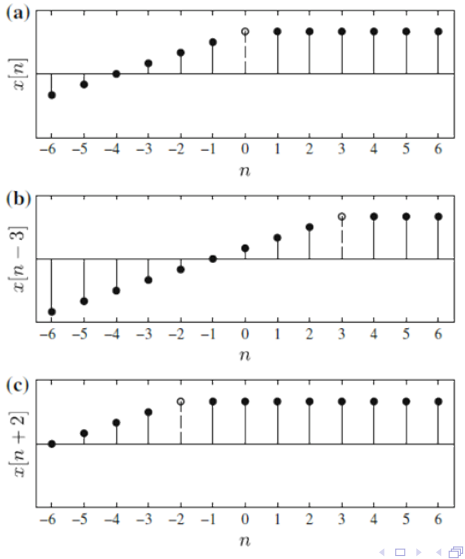
$$x[-n] = \{5,\ 4,\ 3,\ 2,\ \underset{\uparrow}{1}\} \quad x[-n+2] = \{\ 5,\ 4,\ \underset{\uparrow}{3},\ 2,\ 1\}$$

Example: Amplitude Change





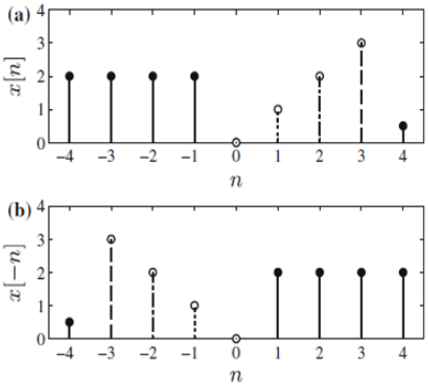
Example: Sequence Shifting



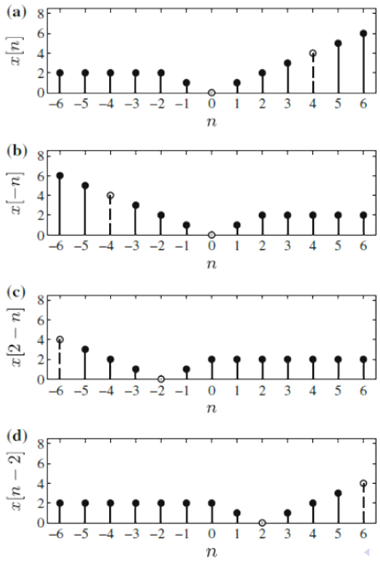
Examples of discrete-time signal translations. **a** Original signal; **b–c** shifted signals. In order to make the transformation easily visible, *empty circles* and *dashed lines* have been used for the sample located originally at $n = 0$. The signal $x[n - 3]$ is delayed with respect to $x[n]$; $x[n + 2]$ is advanced

Example: Reflection or Folding

Example of folding around the origin of discrete times. **a** A signal; **b** its folded version. In order to make the transformation easily visible, *empty circles* and *dotted*, *dash-dotted* and *dashed lines* have been used for some samples

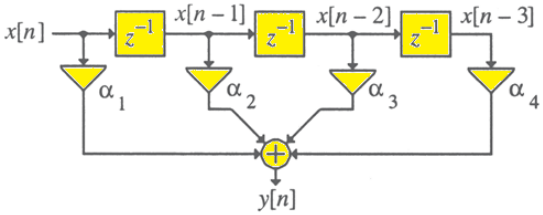


Example: Folding Associated with Translation



Example of folding and translation by 2 time steps. **a** A sequence $x[n]$; **b** the same sequence folded around the origin, $x[-n]$; **c** the folded sequence translated to the left by 2 steps, $x[2-n]$; **d** the sequence $x[n-2]$ obtained by translating $x[n]$ to the right by 2 steps. If we now decided to take this last sequence and fold it, we would once again get $x[2-n]$. In order to make the transformations easily visible, *empty circles* and *dashed lines* have been used for some samples

Combination of Basic Operations



$$y[n] = \alpha_1 x[n] + \alpha_2 x[n-1] + \alpha_3 x[n-2] + \alpha_4 x[n-2]$$

Combination of Basic Operations

- Using basic building blocks, sketch the block diagram representation of the discrete-time system describe the input-output relation

$$y[n] = 0.25y[n-1] + 0.5x[n] + 0.5x[n-1]$$

where $x[n]$ is the input and $y[n]$ is the output of the system.

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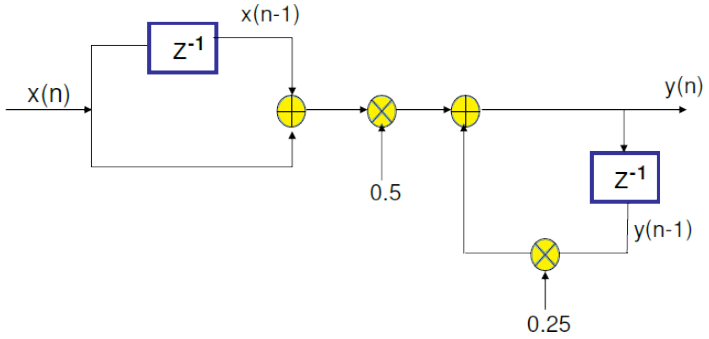


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Up- and Down-Sampling

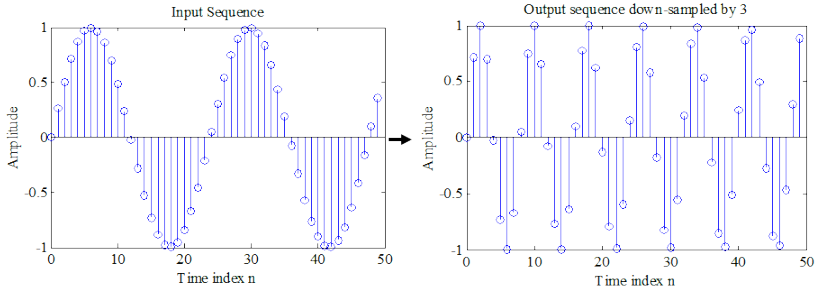
- Certain operations change the **effective sampling rate** of sequences by adding or removing samples
  1. Up-sampling = adding more samples = **interpolation**
  2. Down-sampling = discarding samples = **decimation**
- In **down-sampling** by an integer factor $M > 1$, every $M - th$ samples of the input sequence are kept and $M - 1$ in-between samples are removed:

$$x_d[n] = x[nM]$$



Up- and Down-Sampling

- An example of down-sampling



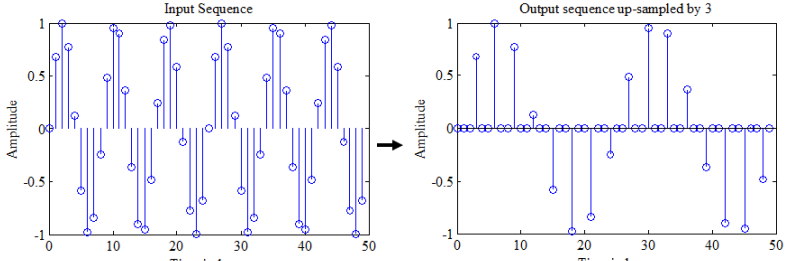
Up- and Down-Sampling

- **Up-sampling** is the converse of down-sampling (not inverse of downsampling!), where $L-1$ zero values are inserted between each pair of original values.

$$x_u[n] = \begin{cases} x[n/L], & n = 0, \pm L, \pm 2L, \ldots \\ 0, & \textit{otherwise} \end{cases}$$

$$x[n] \longrightarrow \boxed{\uparrow L} \longrightarrow x_u[n]$$

- An example of up-sampling



Up- and Down-Sampling

Example

Suppose the original sequence $x[n] = \{\underset{\uparrow}{1},\ 2,\ 3,\ 4,\ 5\}$

Solution

1. *Down scaling* $y[n] = x[2n]$

$$y[n] = x[2n] = \{\underset{\uparrow}{1},\ 3,\ 5,\ \ldots\}$$

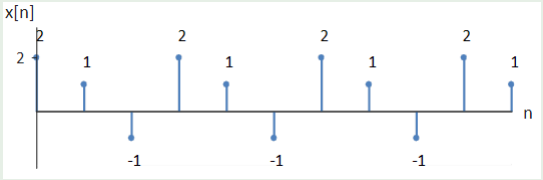
2. *Up scaling or expansion* $y[n] = x[n/2]$

$$[n] = x[n/2] = \{\underset{\uparrow}{1},\ 0,\ 2,\ 0,\ 3.\ 0,\ 4,\ 0,\ 5\}$$

Quiz

Examples

Consider the signal (assume pattern continues for $-\infty < n < \infty$)

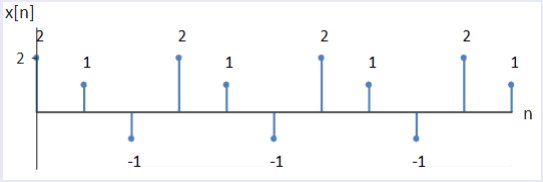


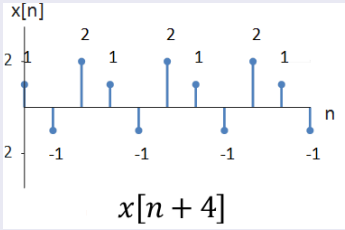
1. Sketch $x[n+4]$
2. Sketch $x[4-n]$

Quiz

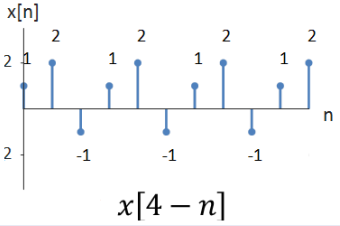
Solution

Consider the signal (assume pattern continues for $-\infty < n < \infty$)





$x[n+4]$



$x[4-n]$

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Complex Sequences

- Two representations of any complex sequence $x[n]$
  1. **Cartesian**: Comprised of in-phase and quadrature phase components

$$x[n] = x_I[n] + j.x_Q[n]$$

  2. **Polar**: Composed of magnitude and phase components

$$x[n] = |x[n]| \exp^{(j.\angle x[n])} = |x[n]| \exp^{(j.\Theta)} = |x[n]| \cos\theta + j|x[n]| \sin\theta$$

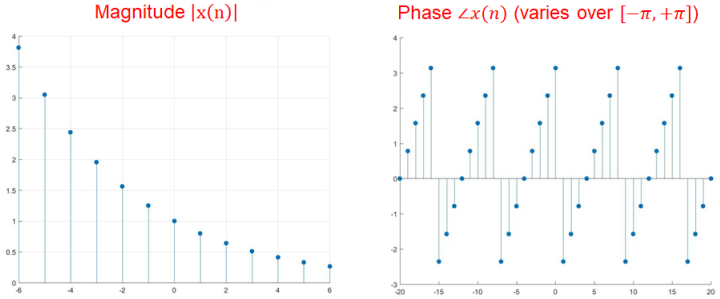
- Relationship between the two representations
  1. Magnitude is given as $|x[n]| = \sqrt[2]{(x_I^2[n] + x_Q^2[n])}$
  2. Phase is given as $\angle x[n] = \tan^{-1}[x_Q[n]/x_I[n]]$

Complex Sequences

Example

When $a = 1$, $x[n] = a^n \exp^{j(2\pi fn+\theta)} = \exp^{j(2\pi fn+\theta)}$ has the following representations

- In-phase/quadrature phase: $x[n] = \cos(2\pi fn + \theta) + j.\sin(2\pi fn + \theta)$
- Magnitude/phase: $|x[n]| = 1$; $\angle x[n] = 2\pi fn + \theta$

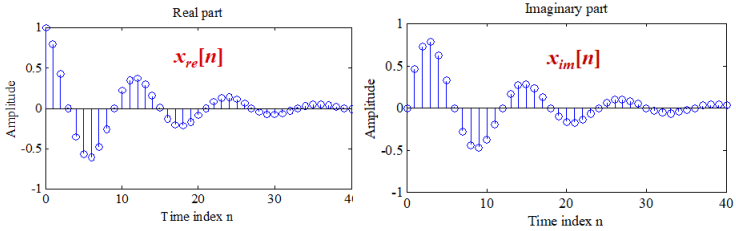


$a = 0.8; f = 0.125$

Complex Sequences

Example

When $x[n] = \exp(-\frac{1}{12} + j\frac{\pi}{6}n)$ has the following representations



$x_{re}[n] = e^{n/12}\cos(\pi n/6)$ $\quad$ $x_{im}[n] = e^{n/12}\sin(\pi n/6)$

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Summary

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- System representation
  - A functional module between an input signal and an output signal
- Operation on a Sequence
  - Addition operation
  - Multiplication operation
  - Product (modulation) operation
  - Time shifting

Any Questions?

